MORPHISMS AGREEING AWAY FROM T AND ON THE COMPLETION ARE EQUAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-faithful-general; the proof environment immediately following it is its own proof

Editorial note. The label lemma-faithful-general names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the /T completion functor of the section and equality of morphisms on completions
- equalizers of morphisms of algebraic spaces: the equalizer E of a and b is an algebraic space and E to X is a monomorphism locally of finite type (spaces-morphisms-lemma-properties-diagonal)
- the factorization lemma of the same chapter (lemma-factor): a morphism f : X to Y factoring through a finite type monomorphism g : Z to Y both away from a closed T and on the completion X_{/T} factors through g globally

Source reference. Original reference 09QM: sources/stacks-restricted/pinned-original-source.tex, source lines 6454–6454. Prerequisite pointer invoked at proof line(s) 6496; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03HK: sources/stacks-chow/tags.txt, source lines 4570–4570. Prerequisite pointer invoked at proof line(s) 6495; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X, W be algebraic spaces over S with X locally Noetherian. Let $T \subset |X|$ be a closed subset. Let $a, b : X \to W$ be morphisms of algebraic spaces over S such that $a|_{X \setminus T} = b|_{X \setminus T}$ and such that $a_{/T} = b_{/T}$ as morphisms $X_{/T} \to W$. Then $a = b$.*

Proof. Let E be the equalizer of a and b. Then E is an algebraic space and $E \to X$ is locally of finite type and a monomorphism, see Morphisms of Spaces, Lemma 03HK. Our assumptions imply we can apply Lemma 09QM to the two morphisms $f = \mathrm{id} : X \to X$ and $g : E \to X$ and the closed subset T of $|X|$. □

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